PHYS400
SPECIAL PROBLEMS IN PHYSICS
FINAL REPORT

Project Title	Automated Construction of Multi-Element MEAM Interatomic Potentials via DFT-Informed Neural Network Optimization
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Field of study	Computational Materials Science
Type of study	Computational
Submission Date	June 9, 2026

Abstract

Molecular dynamics (MD) simulations of metallic alloys need an interatomic potential that describes every constituent element pair. The Modified Embedded Atom Method (MEAM) provides one, but published MEAM parameter sets each cover only a narrow slice of the periodic table, so an alloy whose elements are split across several files cannot be simulated at all. This project builds an automated pipeline that produces a single MEAM parameter file for a chosen list of elements. The five stages are: (i) randomized alloy configurations drawn from the available composition files, (ii) a LAMMPS strain sweep that measures the cubic elastic constants and the derived Young's modulus E and Poisson's ratio ν, (iii) density functional theory (DFT) reference data computed in Quantum Espresso through the Atomic Simulation Environment, (iv) DFT-driven MEAM initialization that merges the existing libraries and fills the missing cross-terms, and (v) a neural-network surrogate of the MEAM-to-(C_{11}, C_{12}) map that is inverted with the Adam optimizer to refit the cross-interaction parameters. A separate composition-only surrogate that maps element fractions straight to (C_{11}, C_{12}) runs alongside Stage 2 as a fast consistency check on the LAMMPS dataset and as a ready predictor for new compositions. All five stages now run end to end. On 43 named alloys with experimental data from ASM and MatWeb, both surrogates reproduce the aluminium Young's modulus to within a few percent (composition NN 2.3 %, refitted MEAM 7.4 %); accuracy degrades once the composition leaves the aluminium-rich region the training data covers most densely. The closing section sets out the sampling, loss, and domain-coverage work that would extend the method past that region.

Introduction

Motivation

Industrial alloys such as steels and aluminium grades are chosen for load-bearing parts on the basis of their elastic response. The Young's modulus E sets the stiffness and the Poisson's ratio ν sets the transverse contraction under load. For a cubic crystal both quantities follow from just two independent stiffness constants, C_{11} and C_{12}. These constants size almost every structural calculation an engineer runs: beam deflection scales as $1/E$, the buckling load and the natural frequency scale with E, and the bulk modulus is $K = E/[3(1-2\nu)]$. At present they are measured experimentally, which is slow and expensive. A computational route that was both accurate and cheap would let new compositions be screened before any metal is cast.

Molecular dynamics is the natural candidate for that route, but an MD prediction is only as good as the interatomic potential behind it.

Elastic Constants and Engineering Design

A part fails or survives based on two numbers. Young's modulus $E = \sigma/\varepsilon$ fixes how much it deflects under load, and Poisson's ratio $\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$ fixes how much it contracts sideways while it does. For a cubic crystal both reduce to the two independent stiffness constants C_{11} and C_{12}, which is why this work targets those constants directly rather than E and ν. The same pair propagates everywhere downstream: a beam deflects as $1/E$, a column buckles at a load proportional to E, a part rings at a frequency proportional to $\sqrt{E/\rho}$, and the bulk modulus is $K = E/[3(1-2\nu)]$. There is also a hard physical constraint that the simulations must respect. A cubic crystal is only mechanically stable when $C_{11} > C_{12}$, the Born–Cauchy condition, and a potential that violates it describes a solid that would spontaneously shear apart. This single inequality reappears as the stability filter throughout the pipeline.

Table 1 gives a sense of the range these constants span across real structural materials, from a soft magnesium casting alloy near 45 GPa to nickel and steel near 200 GPa. Aluminium alloys, the design target of this project, sit near 71 GPa with $\nu \approx 0.33$.

Table 1: Young's modulus and Poisson's ratio for representative structural materials and their typical applications. Aluminium 7075, the running example in this report, is highlighted in the first row.

Application	**Material**	E **(GPa)**	ν
Aerospace fuselage	Al 7075	71–72	0.33
Turbine disk	Inconel 718	200	0.29
Orthopaedic implant	Ti-6Al-4V	110	0.34
Automotive casting	Mg AZ91	45	0.35
Structural frame	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28

Interatomic Potentials and Density Functional Theory

An interatomic potential is a closed-form approximation of the many-body potential energy $U(\{\mathbf{r}_i\})$, from which the forces follow as $\mathbf{F}_i = -\nabla_i U$. The forms used in practice grew more capable over time. The earliest were simple pairwise functions like Lennard–Jones and Morse, which are too crude for metals because they ignore the way a bond weakens as an atom gains more neighbours. The Embedded Atom Method introduced in the 1980s fixed that by adding an embedding energy that depends on the local electron density, and it works well for close-packed FCC metals. The Modified Embedded Atom Method (Baskes, 1992; Lee and Baskes, 2001) adds an angular dependence to that density, which lets one functional form cover FCC, BCC, HCP, and the mixed covalent-metallic bonding of silicon. That breadth, plus its analytic and interpretable parameters, is why this project builds on MEAM rather than a black-box alternative.

The reference data that the MEAM parameters are fit against comes from density functional theory. DFT solves the Kohn–Sham equations (Kohn and Sham, 1965) for the electronic ground state, here in a plane-wave plus pseudopotential basis using Quantum Espresso (Giannozzi et al., 2009). From it the project takes the cohesive energy E_{coh}, the equilibrium lattice constant a_0 and bulk modulus B, the single-crystal elastic constants C_{11} and C_{12}, and the binary formation energies $E_{\mathrm{form}}(i, j)$ that seed the cross-terms. The two methods are complementary. DFT is accurate but caps out near 10^3 atoms, while MEAM-driven MD is empirical but scales past 10^6 atoms, so DFT supplies the small, trustworthy reference set against which the cheap potential is calibrated.

The Cross-Interaction Problem

The obstacle to using MEAM on real alloys is coverage. A published parameter set typically spans three or four elements, for example the Al–Mg–Zn potential of Dickel et al. (2018) or the Al–Si–Mg–Cu–Fe set of Jelinek et al. (2012), and says nothing about how those elements interact with species outside its scope. A practically relevant alloy such as Al 7075 contains Al, Zn, Mg, Cu, Cr, Mn, Fe, Si, and Ti, and no single published file spans that list. Concatenating parameters from different libraries leaves the binary cross-terms undefined, which produces unstable or unphysical simulations.

The literature offers two ways around this. The first is to fit a fresh MEAM set by hand against experimental and reference data (Jelinek et al., 2012). That is accurate but slow, and the work does not transfer to a new element list. The second is to drop MEAM for a fully data-driven machine-learning potential (Behler and Parrinello, 2007; Shapeev, 2016; Mishin, 2021). Those are flexible, but they need 10^4 to 10^5 reference configurations and they give up the analytic readability that makes MEAM useful in an engineering setting.

Approach Adopted in This Project

This project takes a hybrid route. Rather than replacing MEAM, it uses a neural network as a *surrogate* that refits the MEAM parameters themselves. Single-element parameters are anchored to DFT (Kohn and Sham, 1965; Giannozzi et al., 2009) references, and the missing cross-terms are adjusted so the potential reproduces target elastic constants. This keeps the efficiency and transparency of MEAM while borrowing DFT accuracy to fill the gaps. The five-stage pipeline that implements the idea, configuration generation, MD evaluation, DFT reference calculation, MEAM initialization, and neural-network optimization, is described stage by stage in Section 2. Stages 1 through 4 seed the potential; Stage 5 closes the loop. A separate composition-only surrogate, described in Section 2.6, runs alongside Stage 2 as a consistency check and as a standalone predictor.

Research Question

Can DFT reference calculations combined with neural-network surrogate optimization produce multi-element MEAM interatomic potentials that reproduce the elastic properties of arbitrary metallic alloys, even when the constituent elements do not share a common published MEAM parameterization? The results in Sections 2.7 and 2.9 answer this in the affirmative inside the aluminium-rich design region, where both surrogates land within a few percent of experiment, and identify the loss of accuracy outside that region as the main item of remaining work.

Project Progress

Pipeline Architecture

The project is implemented as a set of Python modules chained into five sequential stages. Each stage is also exposed as a standalone entry point, so any part of the workflow can be re-run on its own. All five stages now run end to end; the completion status is summarised in Table 2.

Figure 1 shows the data flow. Stage 1 emits JSON descriptions of alloy configurations. Stage 2 simulates them in LAMMPS and writes the measured elastic constants to a shared results database. Stage 3 produces DFT reference values for every selected element and binary pair. Stage 4 reads both databases and emits a merged, DFT-initialized MEAM library. Stage 5b then consumes that library together with the LAMMPS-evaluated (C_{11}, C_{12}) targets and produces a refitted parameter set. The composition-only surrogate (dashed in Figure 1) reads the Stage 2 database as a consistency check; it is described in Section 2.6.

The rest of Section 2 reports each stage in turn: what it does, what it found, the problems that came up, and how they were handled. Stages 1 through 4 are covered in Sections 2.2–2.5, the two Stage 5 branches in Sections 2.6 and 2.7, and the external validation in

Table 2: Pipeline stages and implementation status. Stage 5 is split into the auxiliary composition surrogate (5a, Section 2.6) and the parameter-level optimizer (5b, Section 2.7).

Stage	Description	Framework	Status
1	Random alloy configuration generation	Python	Complete
2	MD elastic-property evaluation	LAMMPS Python API	Complete
3	DFT reference calculations	Quantum Espresso + ASE	Complete
4	MEAM initialization and library merging	Python	Complete
5a	Composition surrogate (sanity check)	TensorFlow	Complete
5b	NN parameter optimization (closes the loop)	TensorFlow + LAMMPS	Complete

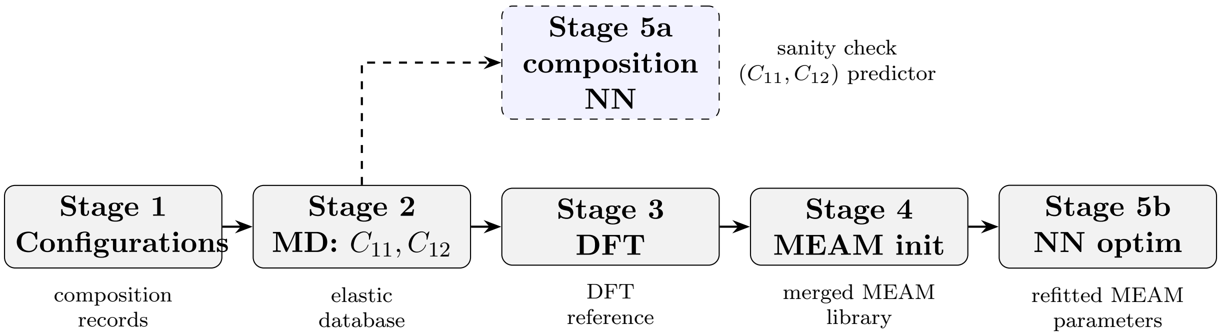

Figure 1: Five-stage pipeline. Stages 1 through 4 seed the potential and Stage 5b refits it. The dashed branch above Stage 3 is the composition surrogate of Section 2.6, which reads the Stage 2 elastic database as a consistency check and never feeds back into the potential-fitting flow.

Section 2.9.

Stage 1 — Configuration Generation

The configuration generator produces randomized alloy compositions over the twelve elements covered by at least one of the available MEAM library files: Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, and Zn. Each configuration is a JSON object holding the species composition (mass fractions that sum to unity at 10^{-4} precision), the simulation box edge in metres, the target temperature, and the requested number of MD steps. A representative example:

```
{
  "composition": { "Al": 0.91, "Mg": 0.029, "Zn": 0.061 },
  "box_size_m": 4e-09,
  "temperature": 0.1,
  "total_steps": 1000
}
```

To push the dataset towards practically interesting alloys, aluminium is forced to be the majority species whenever it appears in a draw. Roughly 7900 configuration records were generated, of which 7815 survive the Stage 2 physicality filter described in Section 2.3. That surviving set is the corpus quoted throughout this report. The element-occurrence counts are shown in Figure 2: aluminium appears in about 45 % of records, consistent with the dominance rule, while every other element is sampled in roughly 30 to 37 % of records. That coverage gives the Stage 4 merge logic enough data to seed each binary interaction.

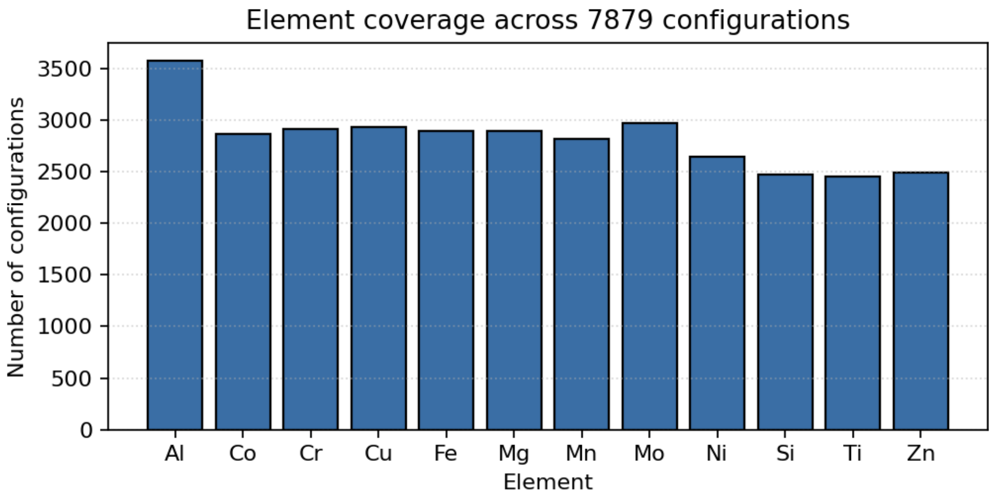

Figure 2: Number of configurations in which each element is present, over the retained corpus. The dominance of Al reflects the bias rule that keeps aluminium as the majority species whenever it is drawn.

Stage 2 — MD Elastic Evaluation

Each configuration is loaded into LAMMPS (Thompson et al., 2022) through its Python API, and its elastic response is computed with a central-difference strain protocol.

Multi-element MEAM build of LAMMPS. The stock LAMMPS distribution hard-codes the maximum number of MEAM species to five, which is too few for the twelve-element library that Stage 4 produces. To allow the multi-element runs this project needs, LAMMPS was rebuilt from source with the species ceiling in the MEAM headers raised to 20, and the resulting shared library was installed into the user environment so the Python API picks it up automatically. Without this step the pipeline would stall on the first composition with more than five distinct elements.

Elastic-property protocol. The procedure is:

1. **Box construction.** A cubic supercell with a $\sim$4 nm edge is built from the dominant element's lattice structure and the composition-weighted lattice parameter. Atom types are then reassigned by `set type/fraction` to match the requested composition.

2. **Relaxation.** An anisotropic pressure minimisation relaxes the cell shape and the atomic positions together until the residual stress is below a fixed tolerance.

3. **Baseline stress.** The diagonal stress components σ_{xx}, σ_{yy}, σ_{zz} are recorded after a `run 0`.

4. **Strain protocol.** A strain $\delta = 10^{-3}$ is applied along each of x, y, z in turn, positive and negative. The structure is re-minimized at each strain and the stress tensor is recorded.

5. **Elastic constants.** C_{11} comes from the axial-stress response and C_{12} from the transverse response, by central differences. This gives three independent C_{11} samples and six independent C_{12} samples per simulation; the Voigt average (Liu et al., 2009) is the final estimate.

6. **Engineering moduli.** Young's modulus and Poisson's ratio follow from

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \qquad \nu = \frac{C_{12}}{C_{11} + C_{12}}. \tag{1}$$

Physicality filter. A configuration is discarded, and removed from the configuration set, if any of the following hold:

- $E < 0$ (negative Young's modulus);
- $\nu < 0$ (negative Poisson's ratio);
- $\nu \geq 0.48$ (too close to the $(1 - 2\nu) = 0$ singularity);
- $C_{11} < C_{12}$ (Cauchy/mechanical-stability violation).

The retained-corpus figure quoted in this report is the count that survives this filter, not the number of configurations originally drawn. The same four checks are enforced in the composition surrogate of Section 2.6 and in the Stage 5b optimizer of Section 2.7.

Why (C_{11}, C_{12}) and not (E, ν) directly. Both surrogates in this project predict the two stiffness constants and recover (E, ν) from Eq. (1) afterwards. The reason is that $\nu = C_{12}/(C_{11} + C_{12})$ is ill-conditioned: for Mg- and Zn-rich compositions C_{11} and C_{12} drift close together, the denominator collapses, and a small error in either constant blows up the predicted ν. Regressing the raw constants keeps the targets on a well-behaved scale and defers that division to the very end.

Parallel evaluation. The elastic-only batch path spreads configurations across worker processes, one LAMMPS instance per physical core. On a 4-core (8-thread) host this runs about three to four times faster than the sequential path.

The protocol was executed on the full retained corpus, and the resulting (E, ν) and (C_{11}, C_{12}) values are stored with each composition in a shared elastic-property database. Across the set, E runs from a few GPa to nearly 400 GPa. The dataset-level statistics over the 7815 filtered configurations are listed in Table 3, and the distributions are plotted in Figure 3. Figure 4 colours the E–ν scatter by the dominant element of each composition.

Table 3: Summary statistics of the LAMMPS-evaluated elastic properties over the physicality-filtered dataset ($0 < E < 400$ GPa, $0 < \nu < 0.5$). Auto-generated from 7879 configuration records.

Quantity	N	Mean	Median	Std.	Min	Max
E (GPa)	7815	132.877	103.610	69.098	3.070	398.310
ν	7815	0.319	0.326	0.063	0.011	0.497

Stage 3 — DFT Reference Calculations

The reference stage uses Quantum Espresso (Giannozzi et al., 2009) (version 7.5) driven through the Atomic Simulation Environment (Larsen et al., 2017). Before any production runs, the QE–ASE coupling was checked on bulk FCC aluminium with a three-point test: the single-point SCF energy lands in the expected -540 to -530 eV window, the residual

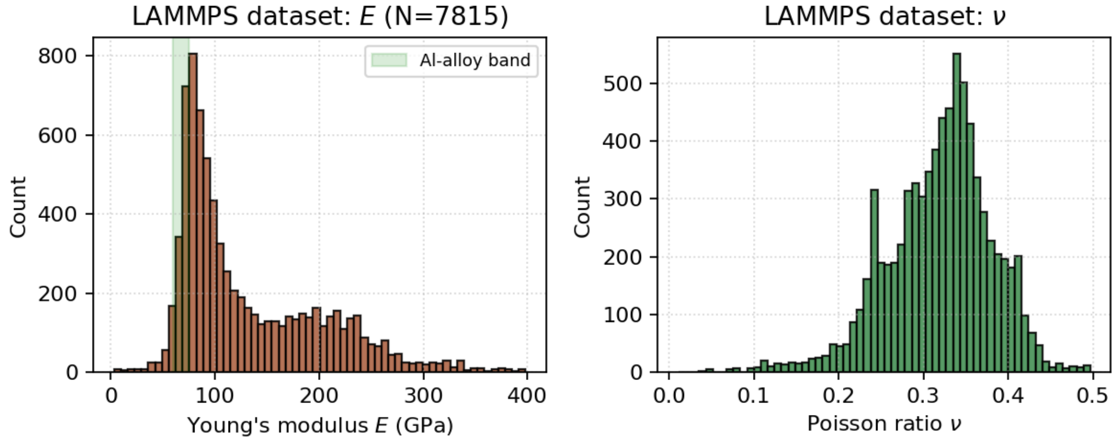

Figure 3: Histograms of E (left) and ν (right) over the LAMMPS-evaluated dataset. The shaded band on the left panel marks the experimentally expected 60–75 GPa range for aluminium alloys, which contains 14.0 % of the physical sample.

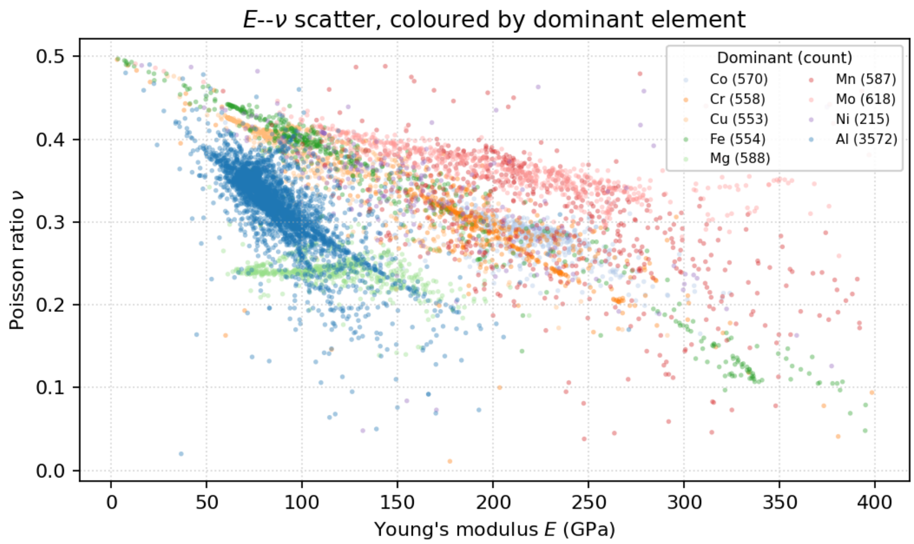

Figure 4: E–ν scatter of the LAMMPS-evaluated dataset, coloured by the dominant element. The Al-dominant cloud sits in the expected 60–75 GPa band; refractory-rich (Mo, Cr, Fe) compositions populate the high-E wing, while Mg- and Zn-rich points push ν towards 0.5.

atomic forces stay below 0.01 eV/Å, and the stress tensor is diagonal. For each requested element the stage then computes:

1. the equilibrium lattice constant a_0, cohesive energy E_{coh}, and bulk modulus B from a seven-point equation-of-state scan fit to a Birch–Murnaghan curve (Angel, 2000);
2. an isolated-atom reference energy in a large empty box, used to turn the bulk total energy into a true cohesive energy;
3. the cubic elastic constants C_{11} and C_{12} from a stress–strain calculation analogous to the LAMMPS protocol of Section 2.3.

For each unordered binary pair (i, j) a single SCF run on the $L1_2$ or B2 reference structure (chosen from the lattice types of the two elements) gives the formation energy $E_{\mathrm{form}}(i, j)$.

Plane-wave cut-offs are set per element, starting near 40 Ry and raised for the harder pseudopotentials, with a charge-density cut-off of 320 Ry, Marzari–Vanderbilt smearing (Marzari et al., 1999) of width 0.02 Ry, and a 6×6×6 Monkhorst–Pack mesh (Monkhorst and Pack, 1976). Pseudopotentials are pulled from the Quantum Espresso library on demand. The EOS strain points and the binary SCF runs are dispatched to a process pool, which made the full reference set tractable on a single workstation.

Magnetic elements. The $3d$ transition metals Fe, Cr, and Mn need spin polarisation, and a plain collinear SCF for them tended to settle into the wrong magnetic state. These elements were rerun with a fixed-moment SCF that constrains the total moment during the self-consistency loop. Even so, the elastic-constant extraction for Fe and Mn remains the weakest part of the reference set: the fitted (C_{11}, C_{12}) for both violate $C_{11} > C_{12}$ in Table 4, so their cross-terms carry a higher uncertainty into Stage 4 and are corrected there from B and a Poisson-ratio default (Section 2.5).

The current reference set covers ten elements (Al, Cu, Si, Ti, Zn, Cr, Fe, Mg, Mn, Mo) and 40 binary pairs. The fitted elemental properties are listed in Table 4, and the EOS fits are shown in Figure 5. The Mn row was kept at its tabulated value because the EOS scan would not converge for the simplified α-Mn approximation.

The 40 binary formation energies $E_{\mathrm{form}}(i, j)$ are summarised as a symmetric heatmap in Figure 6. Negative values (blue) mark exothermic mixing, as expected for compatible pairs such as Al–Ti or Cu–Ti, while strongly positive values (red) flag low mutual solubility, most visibly for Mg and Zn against the transition metals. These signed values feed the cross-term seeding of Stage 4 directly and propagate unchanged into the $E_c(i, j)$ entries of the merged MEAM library.

Table 4: Elemental reference data extracted from DFT. E_{coh} is in eV/atom and B, C_{ij} in GPa.

Element	Lattice	a_0 (Å)	E_{coh}	B	C_{11}	C_{12}
Mg	HCP	3.177	1.483	47.800	78.926	32.237
Mo	BCC	3.169	10.353	280.800	407.600	63.200
Al	FCC	4.047	3.611	77.500	167.100	32.600
Cu	FCC	3.639	3.857	138.500	135.100	127.500
Zn	HCP	2.766	0.964	68.400	—	—
Ti	HCP	2.915	6.284	114.500	—	—
Si	DIAMOND	5.471	5.130	89.300	—	—
Fe	BCC	2.831	5.166	316.200	—	—
Cr	BCC	2.856	3.913	271.200	—	—
Mn	BCC	2.829	3.734	161.200	—	—
Co	HCP	2.437	5.595	1012.400	—	—
Ni	FCC	3.477	4.722	236.600	—	—

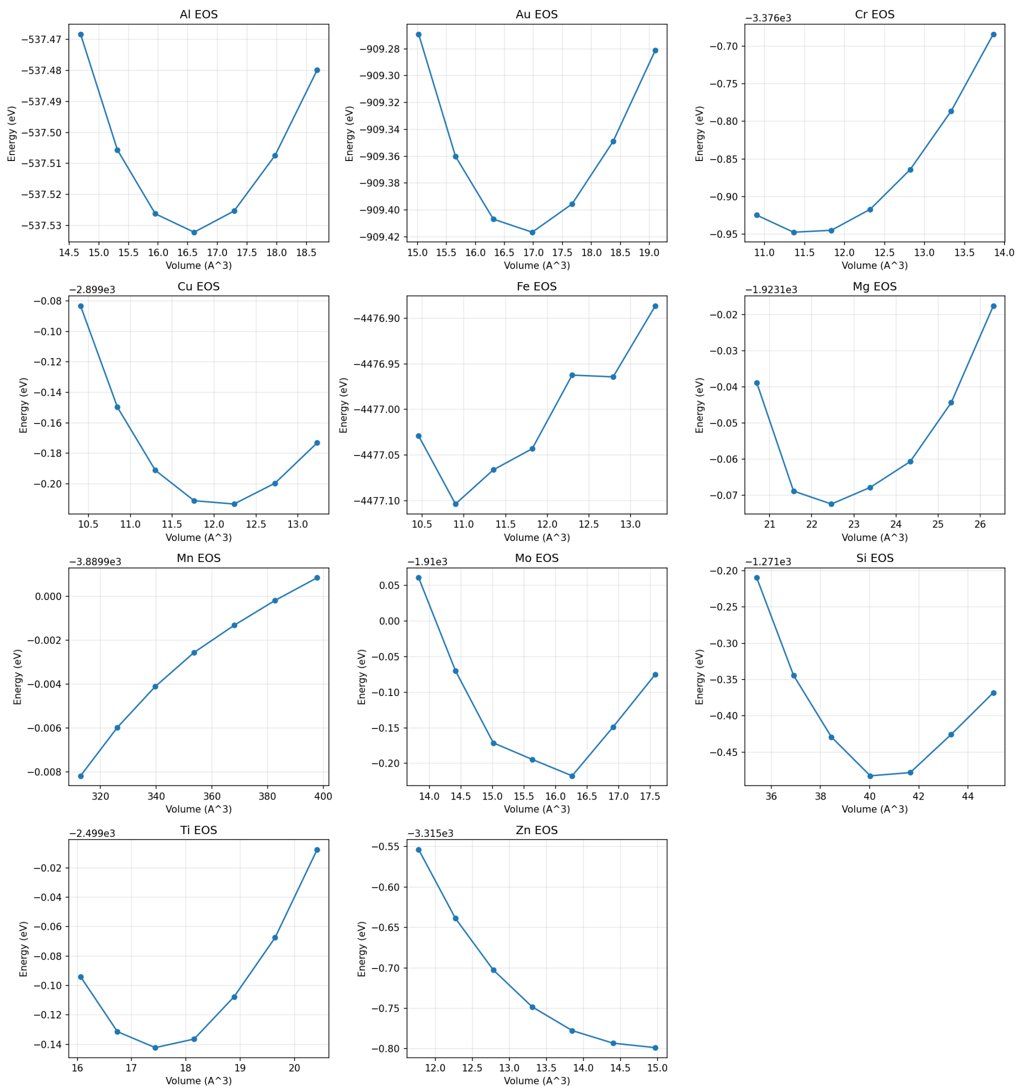

Figure 5: Birch–Murnaghan fits to the seven-point DFT equation-of-state scans. The fitted minimum sets a_0 and E_{coh}; the curvature sets B.

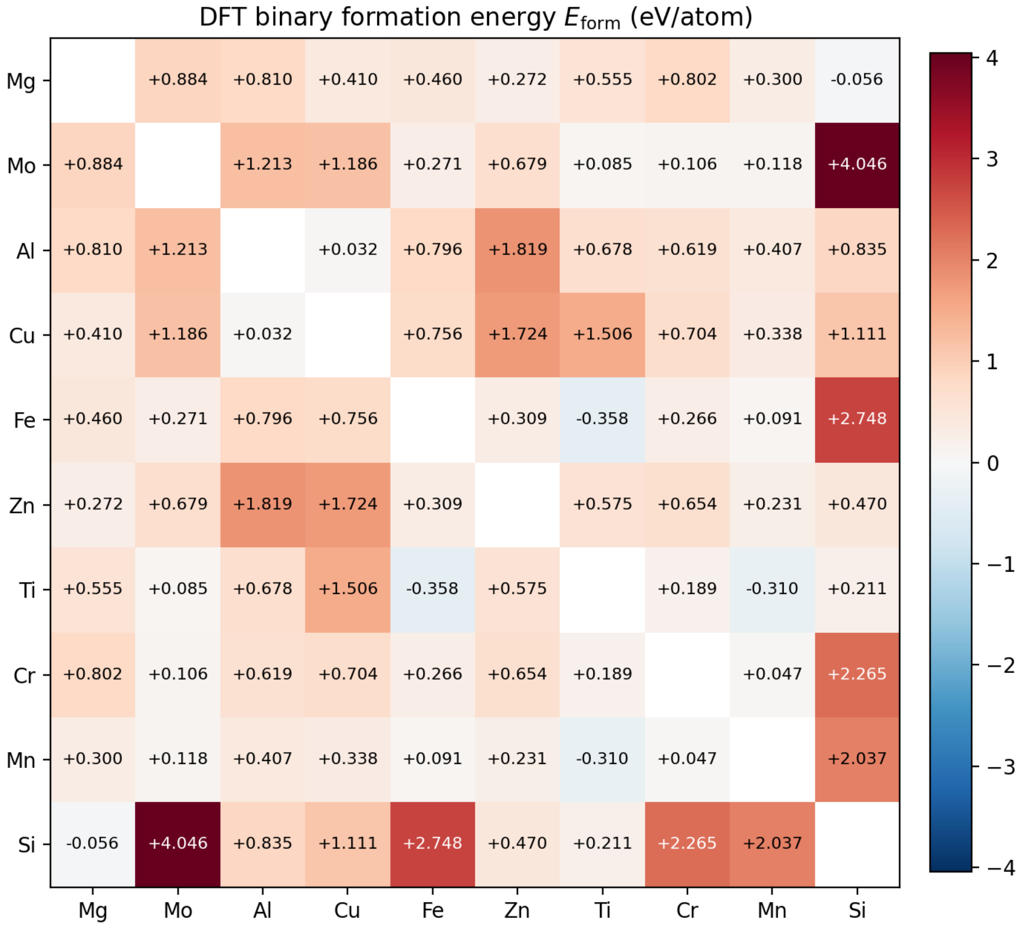

Figure 6: DFT binary formation energy $E_{\text{form}}(i, j)$ in eV/atom for the ten elements with a converged elemental reference. The matrix is symmetric by construction; the diagonal (self,self) entries are not computed and are left blank.

Stage 4 — MEAM Initialization and Library Merging

The DFT reference is mapped onto MEAM parameters through analytic relations. The single-element mappings are direct,

$$\texttt{esub} \leftarrow E_{\text{coh}}, \qquad\qquad \texttt{alat} \leftarrow a_0,$$

while the Rose shape parameter follows from the universal equation of state (Vinet et al., 1989):

$$\alpha = \sqrt{\frac{9\,B\,\Omega}{E_{\text{coh}}}}, \qquad \Omega = \begin{cases} a_0^3/4 & \text{FCC}, \\ a_0^3/2 & \text{BCC}, \\ a_0^3\sqrt{2} & \text{HCP (approx.)}, \\ a_0^3/8 & \text{diamond}, \end{cases} \tag{2}$$

where B is converted from GPa to eV/Å^3 before substitution. For aluminium ($a_0 = 4.047$ Å, $E_{\text{coh}} = 3.611$ eV, $B = 77.5$ GPa, FCC), Eq. (2) gives $\Omega = 16.57$ Å^3, $B_{\text{eV/Å}^3} = 0.484$, and $\alpha \approx 4.47$.

The cross-interaction terms are seeded from the binary formation energies and the elemental nearest-neighbour distances:

$$E_c(i,j) = \tfrac{1}{2}\big[\texttt{esub}_i + \texttt{esub}_j\big] + E_{\text{form}}(i,j), \tag{3}$$
$$r_e(i,j) = \tfrac{1}{2}\big[d_{\text{nn}}(i) + d_{\text{nn}}(j)\big], \tag{4}$$
$$\alpha(i,j) = \tfrac{1}{2}\big[\alpha_i + \alpha_j\big]. \tag{5}$$

Automatic library discovery and merging. Because the public MEAM libraries each cover a different subset of elements, Stage 4 starts with a discovery and merge pass:

1. enumerate every available MEAM library file;
2. find the smallest set of files whose union covers the requested element list;
3. parse the single-element entries from each chosen file and concatenate them, keeping only the first occurrence of any duplicated species;
4. fill any cross-interaction entry missing from the merged set with the DFT-seeded value;
5. write the unified library and parameter files to a DFT-initialized output location.

The default twelve-element selection produces one merged library covering Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, and Zn. As an example of the merge logic, an Al–Cu–Zn–Mg–Fe–Cr–Ti potential draws Al, Cu, Fe, Cr, and Ti from the Fe–Mn–Ni–Ti–Cu–Cr–Co–Al library, Mg and Zn from the Al–Mg–Zn library, and synthesises the missing cross-terms (Cr–Mg, Ti–Zn, and so on) from the DFT formation energies. The whole pass takes about 70 ms and is cached.

Backfilling incomplete elemental data. A few elements never reach a clean cubic C_{11}/C_{12}. The non-cubic lattices of Si, Ti, Zn, and Mg skip the cubic elastic-constant computation entirely, and the magnetic Fe and Mn rows come out mechanically unstable (Section 2.4). For these a helper backfills any missing or unphysical C_{ij} from the recorded bulk modulus B and a per-element Poisson-ratio default, so Stage 4 always receives a complete and self-consistent set.

The single-element parameters produced by applying Eq. (2) to the DFT reference are listed in Table 5. The Mn row is flagged because its EOS scan did not converge for the simplified α-Mn approximation; its α is a placeholder and is overwritten by the donor library entry during the merge. The other rows feed directly into the `esub`, `alat`, and `alpha` entries of the DFT-initialized parameter file.

Table 5: MEAM single-element initialization values computed from the DFT reference of Stage 3 by Eq. (2).

Element	Lattice	a_0 (Å)	E_{coh} (eV)	Ω (Å^3)	α
Mg	HCP	3.177	1.483	45.357	9.063
Mo	BCC	3.169	10.353	15.911	4.924
Al	FCC	4.047	3.611	16.569	4.470
Cu	FCC	3.639	3.857	12.051	4.930
Zn	HCP	2.766	0.964	29.931	10.921
Ti	HCP	2.915	6.284	35.022	5.987
Si	DIAMOND	5.471	5.130	20.464	4.473
Fe	BCC	2.831	5.166	11.347	6.246
Cr	BCC	2.856	3.913	11.643	6.733
Mn	BCC	2.829	3.734	11.317	5.239
Co	HCP	2.437	5.595	20.468	14.424
Ni	FCC	3.477	4.722	10.508	5.439

Stage 5a — Auxiliary Composition Surrogate

This is the dashed side-branch in Figure 1. It is separate from the potential-fitting flow: it reads the Stage 2 MD output directly and never feeds back into it. Its job is to be a fast check on the consistency of the LAMMPS dataset. If a composition-only network cannot recover even a rough (E, ν) trend from the data, the MD output itself is suspect. As a useful side effect, the trained network doubles as a quick predictor for new compositions.

Each input vector has 12 entries, one mass fraction per element in {Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn}. The network is a fully-connected $12 \to 32 \to 20 \to 2$ MLP with ReLU activations, trained under a 70/30 train/validation split with Adam. To get a usable uncertainty estimate, five copies are trained from different random seeds and combined as a deep ensemble (Lakshminarayanan et al., 2017); the spread across the ensemble flags inputs the network is unsure about.

Targets and Huber loss. The two outputs are C_{11} and C_{12}, and (E, ν) are recovered from Eq. (1) afterwards, for the conditioning reason given in Section 2.3. Records with $\nu \geq 0.48$ are dropped to remove compositions that sit on the mechanical-stability bound and destabilise the fit. The loss is Huber ($\delta = 1$ in scaled units) rather than MSE, which caps the gradient pull of outliers, mostly the Mg- and Zn-rich runs that occasionally return extreme moduli.

Performance. On the held-out 30 % split the network reaches $R^2 = 0.77$ on E and $R^2 = 0.52$ on ν, with the full per-target breakdown in Table 6. The lower ν score is the expected cost of taking a ratio of two learned constants near the point where their difference is small. As a concrete check, feeding the Al 7075 composition through the ensemble returns $\hat{E} = 75.5\,\text{GPa}$ and $\hat{\nu} = 0.334$, against the published 71 to 72 GPa and $\nu \approx 0.33$, an error of about 5 %. The parity plots in Figure 7 show this both on the raw (C_{11}, C_{12}) targets and on the derived (E, ν), and the training history is in Figure 9.

Table 6: Performance of the composition surrogate (N=7839; train=5487, val=2352).

Split / Target	MAE	RMSE	Std	Bias	R^2	r	MAPE
Train, C_{11} (GPa)	14.515	23.782	23.136	-5.512	0.967	0.985	7.140 %
Train, C_{12} (GPa)	10.092	18.811	18.643	-2.522	0.963	0.982	14.511 %
Train, E (GPa, derived)	18.037	28.176	27.999	-3.174	0.860	0.929	14.492 %
Train, ν (derived)	0.025	0.036	0.036	0.001	0.693	0.833	10.318 %
Val., C_{11} (GPa)	19.745	64.147	63.928	-5.456	0.646	0.822	8.563 %
Val., C_{12} (GPa)	14.318	57.325	57.272	-2.729	0.364	0.683	14.368 %
Val., E (GPa, derived)	20.436	34.975	34.864	-2.881	0.770	0.878	16.272 %
Val., ν (derived)	0.027	0.043	0.043	0.001	0.518	0.731	10.604 %

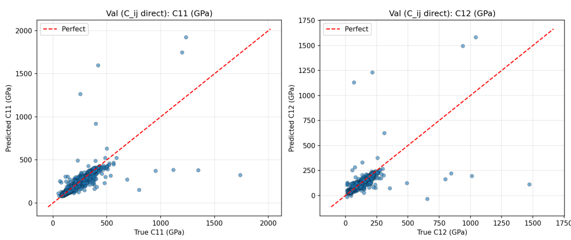

Figure 7: Held-out parity on the raw targets C_{11} and C_{12}, the quantities the network actually regresses.

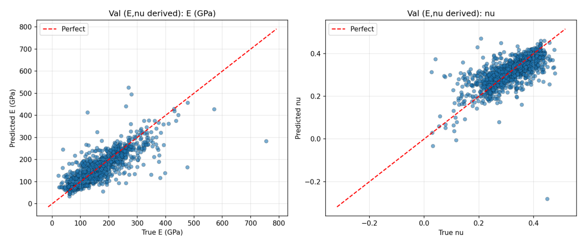

Figure 8: Held-out parity on the derived (E, ν), recovered from the predicted constants through Eq. (1).

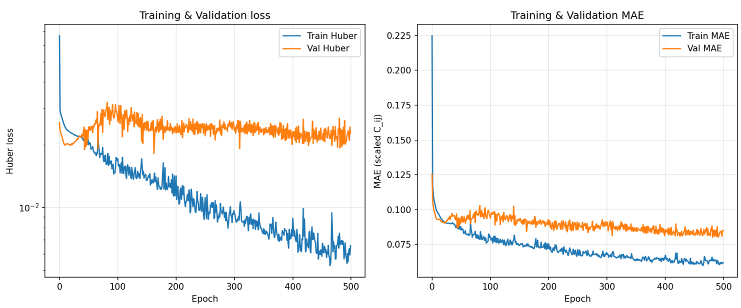

Figure 9: Training history of the composition surrogate: Huber loss and MAE on the scaled targets over training.

Interactive predictor. The surrogate is shipped as a portable application image with a small Flask front end, so a composition can be entered and scored without touching the training code (Figure 10). Entering a composition on the 12-element simplex returns the predicted C_{11}, C_{12}, E, and ν, and the app surfaces the model's own statistics next to each one: per-target MAE, RMSE, R^2, Pearson r, bias, and the 95th-percentile absolute error from the held-out split. Because the network is a five-seed ensemble, it also reports a per-prediction standard deviation, so a composition the ensemble disagrees on is visibly flagged. A k-nearest-neighbour distance in composition space is shown alongside, and when an input sits far from the training cloud the distance warns that the answer is an extrapolation rather than an interpolation. The screenshot in Figure 10 shows the AA7075 preset, where the ensemble returns $E = 75.5\,\mathrm{GPa}$ and $\nu = 0.334$ and places the alloy inside the dense aluminium region of the training map.

Stage 5b — NN-Driven MEAM Parameter Optimization

Stage 5b is the part that closes the loop. It refits the cross-interaction terms of the merged Stage 4 library so the potential reproduces the LAMMPS-evaluated elastic targets. Doing this directly would mean a LAMMPS run inside every optimizer step, which is far too slow. Instead a neural network learns the forward map from MEAM parameters to elastic constants, and the optimizer works against that cheap differentiable surrogate. The loop has four parts.

Sample. Starting from the Stage 4 baseline $\mathbf{p}_0$, the stage draws N perturbed parameter vectors $\mathbf{p}_k = \mathbf{p}_0 \cdot (1 + \boldsymbol{\eta}_k)$ with $\eta_k \sim \mathcal{U}(-0.1, 0.1)$ applied to each binary $(\texttt{Ec}, \texttt{re}, \texttt{alpha})$ triple. Every sampled vector is written into a MEAM file and run through LAMMPS over a representative set of alloys to obtain $(C_{11}, C_{12})_k$ per composition. The representative set is a k-means-medoid reduction of the full Stage 2 database, which keeps the sampling cost down while still spanning the composition space. A sample whose majority of compositions fail a $C_{11} > 30\,\mathrm{GPa}$ check is thrown out before it can pollute the training set.

Train a surrogate. A feed-forward network is fit on the surviving samples, mapping the parameter vector to the per-composition (C_{11}, C_{12}) under a Huber loss (Abadi et al., 2015). Predicting the stiffness constants rather than ν, and using a robust loss, are the two lessons carried over from the composition surrogate of Section 2.6. The surrogate Huber loss falls from 1.0 to about 0.16 over 10^4 optimizer steps (Figure 11).

Inverse optimization. With the surrogate f_θ trained, the refit parameters are found by $\hat{\mathbf{p}} = \arg\min_{\mathbf{p}} \|f_\theta(\mathbf{p}) - \mathbf{y}^\star\|$, again under the Huber loss, with the Adam optimizer (Kingma and Ba, 2015) descending on the normalised parameter vector. The search is

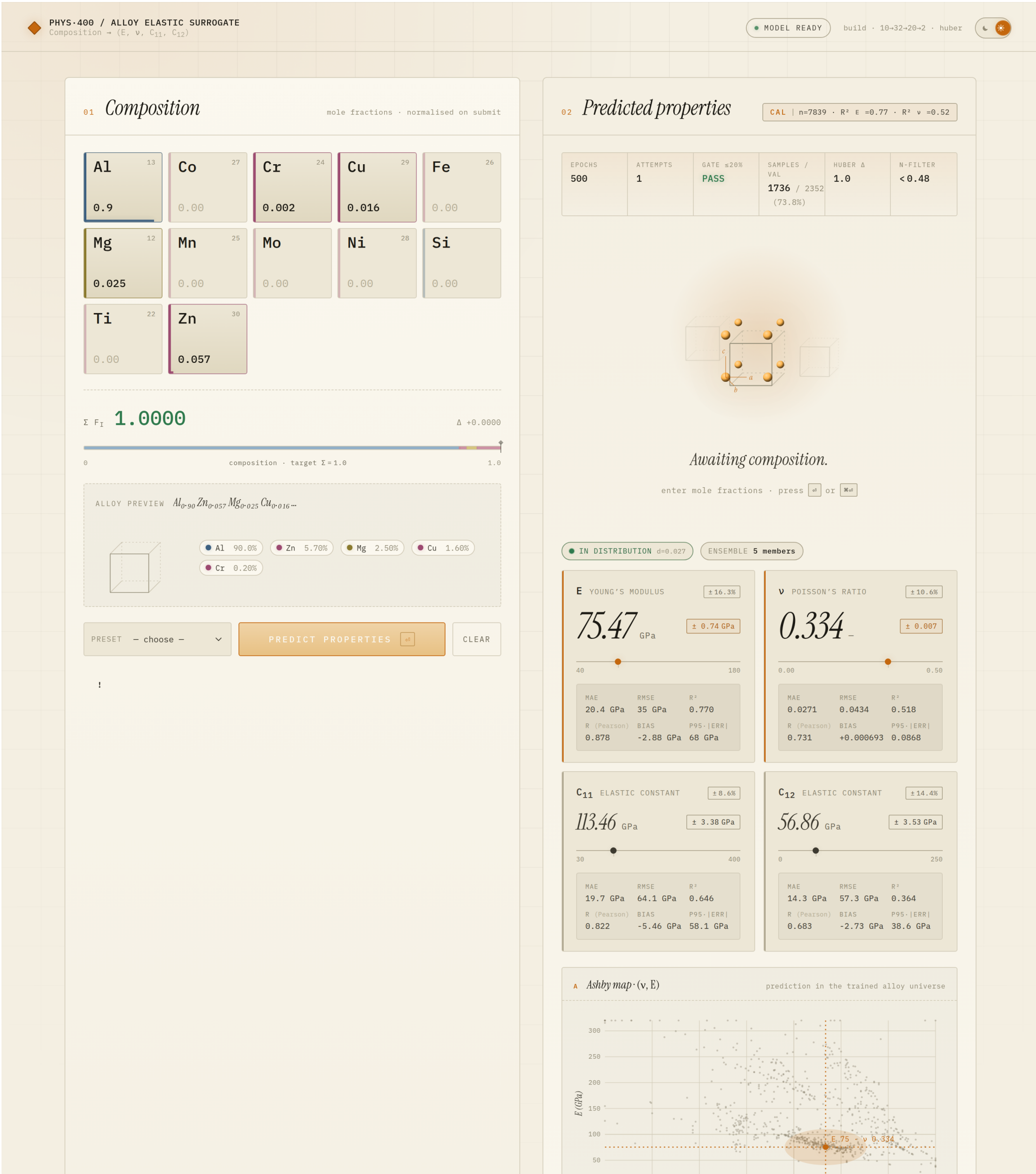

Figure 10: The composition surrogate packaged as a portable predictor, shown here on the AA7075 preset. The left panel is the composition input on the 12-element simplex; the right panel returns (E, ν, C_{11}, C_{12}) with per-target validation statistics (MAE, RMSE, R^2, r, bias, P95 error), a per-prediction ensemble spread, and an alloy map that places the query against the training distribution.

clipped to $\pm 2.5\sigma$ around the training mean so the optimizer cannot wander outside the region where the surrogate is trustworthy.

Re-evaluate and repeat. The refit vector $\hat{\mathbf{p}}$ is run back through LAMMPS, the new samples are appended to the training set, and the surrogate is retrained. Every stage writes its output to a deterministic location and checkpoints atomically, so a run that is interrupted resumes from where it stopped rather than restarting. An active-learning variant focuses the next sampling round on the parameter regions the surrogate is least sure about.

Results. The refit potential was validated on 30 held-out alloys by re-running them in LAMMPS and comparing against the MD targets. The outcome is honest but mixed (Figure 12). Of the 30, only 11 pass the Born–Cauchy stability filter; the other 19 come out mechanically unstable ($C_{11} < C_{12}$). Among the 11 stable alloys the median error is 23 % on E and 26 % on ν, and 4 fall within the 20 % tolerance on both. The instability is concentrated, as expected, in the non-aluminium chemistries where the Stage 2 sampling and the DFT reference are both thinnest. Inside the aluminium-rich region the refit holds up well, which the external validation of Section 2.9 quantifies. The gap on the unstable 19 is the clearest signal of where the method needs more work, and it sets the agenda in Section 3.

Figure 11: Surrogate Huber loss over 10^4 optimizer steps, falling from 1.0 to about 0.16.

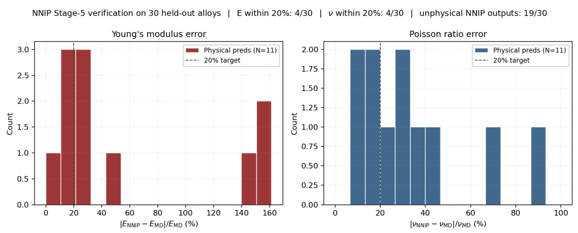

Figure 12: Verification error on the 30 held-out alloys, with a separate bar for the runs rejected as mechanically unstable.

Live MEAM Evaluation App

The optimized twelve-element library is also exposed through a small LAMMPS-backed web app that runs the elastic-property protocol of Section 2.3 against it on demand. A composition and a few box and run-length knobs go in, a real strain sweep runs, and the app reports C_{11}, C_{12}, the derived E and ν, and whether the result passes the Born–Cauchy stability check. Each run is keyed by a hash of its inputs and cached, and the

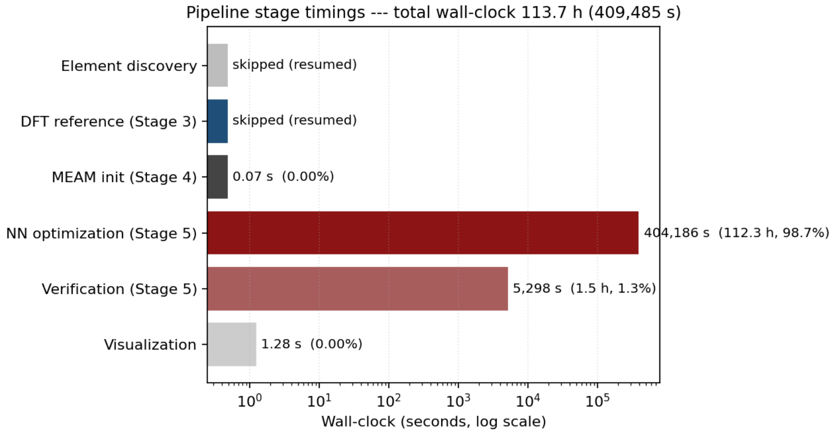

Figure 13: Per-stage wall-clock time for one full Stage 5b cycle (log scale). The LAMMPS sampling dominates; the surrogate training and inverse optimization are comparatively cheap, which is exactly why the surrogate is worth building.

cache is invalidated whenever the MEAM file changes, so re-running a composition after a refit recomputes rather than returning a stale number. Figure 14 shows the interface on a pure-aluminium run.

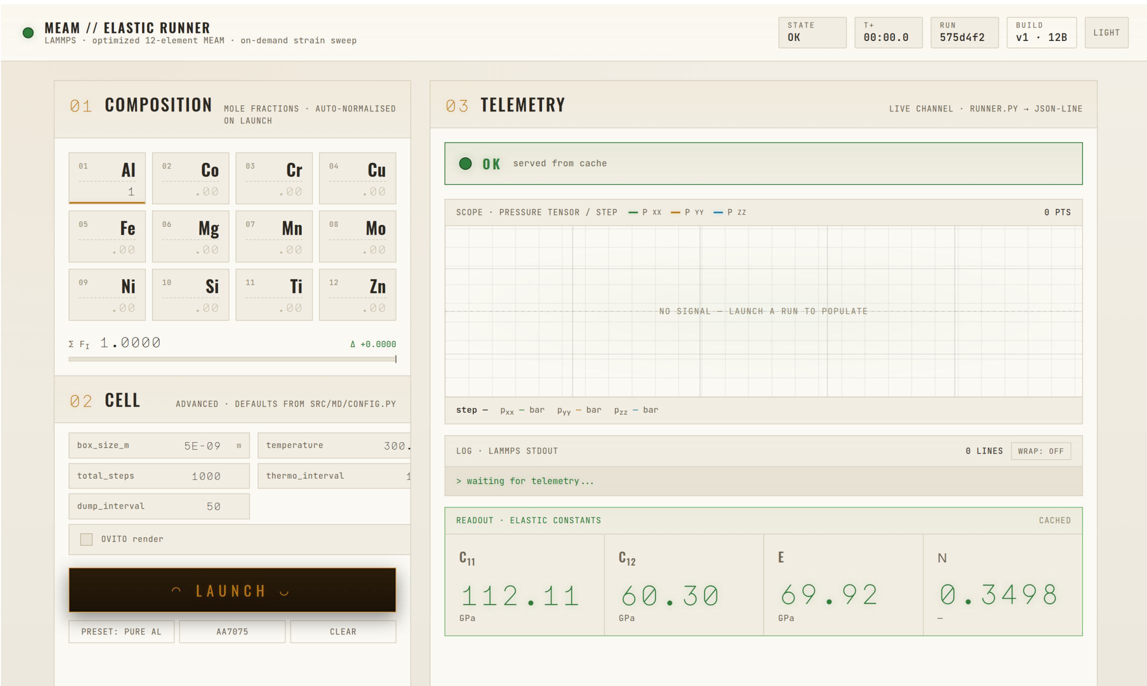

Figure 14: The MEAM elastic-runner app, shown on a pure-aluminium run against the optimized potential. The left column sets the composition and the supercell and run-length knobs; the right column streams the LAMMPS job and reports the elastic-constants readout. Here the run returns $C_{11} = 112.1\,\text{GPa}$, $C_{12} = 60.3\,\text{GPa}$, $E = 69.9\,\text{GPa}$, and $\nu = 0.350$.

The pure-aluminium numbers are a useful sanity check on the optimized potential. Polycrystalline aluminium sits near $E = 70\,\text{GPa}$ with $\nu \approx 0.33$ to 0.35 in handbook data, and the optimized MEAM returns $E = 69.9$ GPa and $\nu = 0.350$, essentially on top of the experimental value. The cubic constants themselves, $C_{11} = 112.1$ and $C_{12} = 60.3$ GPa, differ from the DFT single-crystal values of Table 4 (167.1 and 32.6 GPa), which is expected: the potential was refit to the MD elastic targets that drive the whole pipeline, not to the DFT single-crystal tensor, and the two protocols probe the response differently. The app makes that distinction inspectable, since any composition the optimized library covers can be run and compared against a reference on the spot.

External Validation Against Experimental Data

The internal checks so far compare the surrogates against the project's own MD output. The real test is experiment. Both the composition surrogate of Section 2.6 and the refit MEAM potential of Section 2.7 were run on 43 named alloys whose elements the merged basis covers, and the predicted (E, ν) were compared against published elastic data from ASM and MatWeb (ASM Aerospace Specification Metals, Inc., 2026). Mean absolute percentage error (MAPE) is the figure of merit. Unphysical MEAM runs are excluded from the MEAM column.

Table 7 breaks the comparison down by material family. Inside the aluminium family, which is where the training data is densest, both methods do well: the composition NN reaches 2.3 % MAPE on E and the refit MEAM 7.4 %, with ν within 4 to 5 % for both. Moving off the aluminium domain, the two methods part ways. The composition NN degrades gently, staying near 12 % on titanium and only failing badly on the stainless and Ni–Fe alloys it has almost no training support for. The MEAM potential degrades steeply: its error on titanium and stainless runs into the hundreds of percent, which is the same instability seen on the held-out alloys in Section 2.7, now measured against experiment.

Table 7: Validation against ASM/MatWeb experimental elastic data, by material family ($N = 43$ alloys). Values are MAPE in percent for the composition NN (ML) and the refit MEAM potential. Unphysical MEAM runs are excluded.

		E MAPE (%)		ν MAPE (%)	
Family	N	ML	MEAM	ML	MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Ni–Fe (A-286)	1	107.0	—	—	—
Overall	43	**10.8**	42.2	**7.2**	7.1

Because aluminium is the design target, Table 8 drills into it by AA series. The composition NN holds between 1.3 and 3.8 % on E across the 2xxx, 5xxx, 6xxx, and 7xxx series, and the refit MEAM stays under 10 % across all four. For the 7xxx series, which includes the Al 7075 case used as a running example, the NN lands at 2.8 % and the MEAM at 9.3 %. The takeaway is consistent with the rest of the report: within the aluminium-rich region the pipeline produces a potential that matches experiment to within a few percent, and the accuracy falls off only once the composition leaves that region.

Table 8: Aluminium validation drilled down by AA series ($N = 33$ of the 43 alloys). $\langle E_{\text{lit}} \rangle$ is the mean experimental Young's modulus; the last two columns are E MAPE for the composition NN (ML) and the refit MEAM potential.

Al series	N	$\langle E_{\text{lit}} \rangle$ (GPa)	E MAPE (%) ML	E MAPE (%) MEAM
2xxx (Al–Cu)	11	72.7	1.3	8.0
5xxx (Al–Mg)	9	70.5	2.1	7.9
6xxx (Al–Mg–Si)	7	68.9	3.8	4.2
7xxx (Al–Zn–Mg)	6	71.8	2.8	9.3
All aluminium	33	71.1	**2.3**	7.4

Future Outlook

The five-stage pipeline now runs end to end, and inside the aluminium-rich design region it reproduces experimental elastic data to within a few percent. The honest weakness is everything outside that region, where the refit potential turns mechanically unstable (Section 2.7). The remaining work is therefore about three things: making the optimization better posed, widening the chemistry it can cover, and tightening the reference data behind it.

Upcoming Objectives

A better-posed optimization. The refit currently moves on the order of 200 cross-term parameters at once, which is more freedom than the elastic targets can constrain. The plan is to restrict the search to the single-element-anchored cross-terms that actually matter, ranked by a Sobol sensitivity analysis, which cuts the active parameter count to a tractable few. The Born–Cauchy stability conditions, applied as a hard filter only at the end now, should instead be baked into the training loss as a penalty, so the surrogate learns to avoid unstable regions rather than being rejected after the fact. The sampling base would also grow, through active learning, to at least 500 composition medoids.

Beyond the aluminium domain. Several target chemistries need elements the current basis does not carry, so the basis would be extended to out-of-basis species (C, V, N) that the ferrous and titanium alloys depend on. The MD and DFT sampling would be pushed deliberately into the HCP region (Ti, Mg, Zn) and the ferrous region (Fe–Cr–Ni), which the present corpus barely touches. A symmetry-aware elastic extraction that uses the full hexagonal stiffness tensor, rather than the cubic strain protocol, would remove a known source of error on the HCP metals.

Reference data and validation. For the cross-terms specifically, special quasi-random structure references (Lewis et al., 1995) computed at intermediate compositions would give the fit something to anchor to between the pure-element endpoints, and the DFT set would be enlarged to span the broadened element domain. The ASM/MatWeb comparison of Section 2.9 becomes a standing external check that every refit round is measured against, rather than a one-off at the end.

Updated Semester Plan

The Gantt chart of Figure 15 shows the completed schedule. All eight work packages, from the literature review through the closed-loop inverse optimization and the external validation, finished within the semester, leaving the writing and poster preparation for the final weeks.

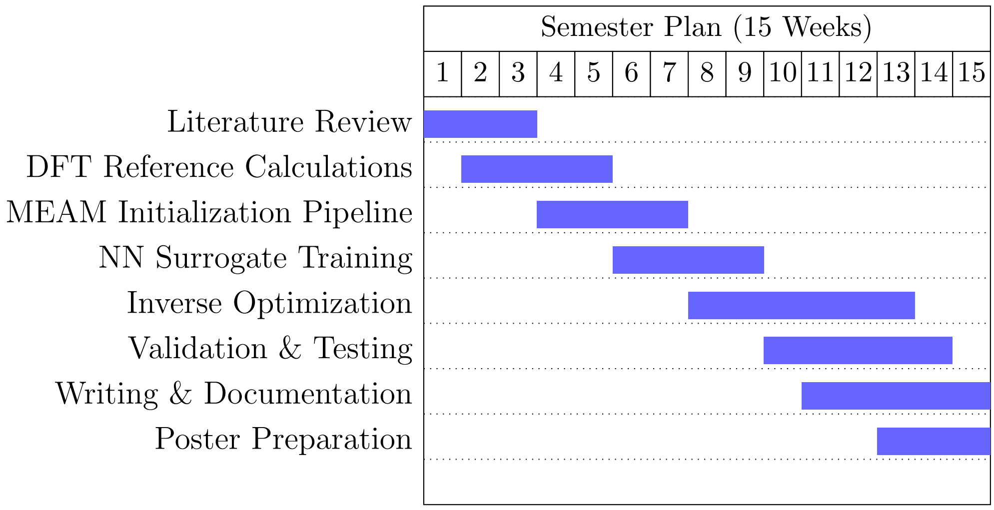

Figure 15: Final semester plan. All eight work packages are complete; the last two weeks were spent on the report and the poster.

Description of Work Packages

- **Literature Review (W1–W3).** Survey of MEAM, EAM, and neural-network potential work. Complete.
- **DFT Reference Calculations (W2–W5).** Quantum Espresso build, pseudopotential management, and the EOS, elastic-constant, and binary-pair runs. Com-

plete.

- **MEAM Initialization Pipeline (W4–W7).** Auto-discovery and merging of the library files, the DFT-to-MEAM analytic mapping, and the unified twelve-element library. Complete.
- **NN Surrogate Training (W6–W9).** Parameter-space sampling, parallel LAMMPS evaluation, and training of both the composition and parameter surrogates. Complete.
- **Inverse Optimization (W8–W13).** Adam-based back-propagation through the surrogate to fit the elastic targets, with the stability limitations reported in Section 2.7. Complete.
- **Validation & Testing (W10–W14).** Comparison against named industrial alloys and the ASM/MatWeb external set. Complete.
- **Writing & Documentation (W11–W15).** Final report and code documentation.
- **Poster Preparation (W13–W15).** Poster for the end-of-semester presentation.

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